Reviewer and editor comments are in italics. Our replies are in plain font.

Associate Editor – comments:

Many thanks for your revision; a small number of points remain to be dealt with.

While your response letter says that you separated pixel sampling from effective optical resolution, it seems that the new version of the manuscript does not (from seeing L. 308- 313 + Figure S1 caption we understand that the stage-micrometer calibration is the resolution measurement). Please address this in your revision.

We have updated the manuscript on line 303 and in Figure S1 to include the minimum feature size we are able to distinguish. We believe this addresses this comment, but please let us know if are misunderstanding the request.

Secondly there is a possible change that will add value to the approach, i.e. by turning the scale bars into a verified measurement. Since ring-width measurement is built around a stitched mosaic preserving distance along the core across a hundred+ frames, you could report the measured vs. true length over the longest span available in that scan (or for a core of known length) especially because you have already scanned a caliper with millimetre graduations (see Table S3).

We thank the editor for the suggestion to add a verifiable measurement to compare the digital sample to the original. Comparing measurements in two different mediums requires precise control points to indicate the start and end points of the measurements. To provide consistency across samples, we designed a rudimentary 3D printed tool to firmly hold two ultra-fine tipped felt pens (Sharpie's) at a fixed distance from one another. We then marked each representative sample with two dots. The distance between the dots on the cores were roughly 101.05 mm. We added Table S5 to detail the control point's physical distance, pixel distance, and overall sample length. We left Table S4 without the control points to keep the breadth of representative scans available for readers.

The images of Figures 3, 4, 5, S1 are a bit blurry. Maybe it has to do with the import, the actual rendering of the pixel limits of the system and/or how it is handled in the pdf generated on ScholarOne but please check whether your resolution is high enough to help with readability.

We thank the editor for noting the blurriness of the figures. We have compared the ScholarOne pdf with the pdf we created and submitted and there is a noticeable blurriness on the ScholarOne pdf. We now additionally provide the figures with our submission as separate files so they can hopefully be viewed in the full resolution. We are not aware of any way to improve the ScholarOne pdf, but happy to make any specific adjustments if requested.

We are missing a detailed figure of the electronic diagram and the basics indications on how to build it. Figure 1 is fine for showing the final product in real life but we need a "full skeleton view" that can be the main part of figure 1. To stay within limits of

the number of words for this kind of paper you can set a detailed description in Suppl. mat. on how to build a TIM, what to do and not to do, the fail safes for the electronics + plus add a table describing each element of TIM, where to get it, the current costs (the paper describes the total costs but we do not have any itemized), etc.

We have added a new section on how to build TIM between line 386 and line 398. This section includes references to Table S1 of the components we bought to build this version of TIM, Figure S2 showing the electrical connections, and a skeleton view of the system in Figure 1. While we were developing TIM, the company which designed and sold the three axis motion system went out of business, which we expect will continue to happen, depending on when new users build TIM. Thus, we have emphasized that new adapters for the camera and computer will need to be designed to adapt to a new machine kit, and note that we will address reasonable requests for help through GitHub.

Our GitHub README.md file has an assembly view image of all the parts in 'exploded' view to reference.

Illumination is also not described anywhere in the Methods, yet it determines contrast and colour rendering and a lab rebuilding TIM will need to know what to buy. Please specify the light source and its mounting. MEE usually asks for manufacturer name, city and country for named equipment (e.g Raspberry Pi, Seeed).

We have added a description of the ring microscope LED light which mounts to the camera lens on line 115.

Regarding the 2.5 GB TIFF limit, tifffile and libtiff both write BigTIFF, which could remove the 4 GB ceiling and would let cookie scans open in ImageJ rather than only through custom scripts you are mentioning in your text.

We thank the editor for the suggestion of saving large scans to a BigTIFF. We changed the code to save the large final stitches as BigTiff's instead of dat files to allow for people to convert the file as they need for their analysis and have updated the manuscript to reflect this on line 324. With this change TIM can successfully save large images up to 12 GB.

We found ImageJ was unable to open the 12GB image that we scanned (76,000 pixels x 75,000 pixels), despite our troubleshooting efforts. ImageJ plugins such as BigDataViewer require a different image file format than BigTiff. We were unsuccessful in our attempts to convert the files to the BigDataViewer format. Downscaled images of the 12 GB full resolution stitch can still be viewed. If a sample requires analysis at full resolution for this large of a file, it will require further investigation by an image analysis expert. We have not required such large files for the analyses we have ran so far.

Talking about custom scripts, I could not find any code/scripts online on the zenodo repository, only Tiff images. In figure 1, we can read that "the 3D printed parts are available for download on the NIH 3D printing repository (HYPERLINK REMOVED)

and the software is open source, with instructions to recreate this tool available on GitHub (HYPERLINK REMOVED)." I thank the authors for ensuring anonymity for the review process but if we are not able to assess all those in the review it is a pb. By going to the NIH 3D website though and typing "TIM" I was able to find the printing instructions but also the name of the first author. Yet I could not find the code and custom scripts references written in the ms; MEE needs to review the codes during the review process, and this will be crucial in order to help with repeatability.

We apologize for failing to add the anonymized code in the revision submission. We have added a zipped file of the anonymized code to this submission. We would like the link to the active GitHub repository included in the final version of the manuscript if accepted (and this is allowed) to promote access to the active repository where users can post questions, see updates, etc..

In Figure 6, the caption states that samples range from a 3 mm × 220 mm core to a 75 mm × 56 mm cookie; yet panel (a) extends to 300 mm and panel (b) stops near 1,600 mm², all short of the 4,200 mm² such a cookie would occupy. Panel (b) also joins four cookies with lines, yet four points cannot be the base of a relationship; with 80 cookies scanned, more points should be available too. Please also state the overlap setting used for the plotted runs.

We thank the editor for finding this mistake. We now include more cookie scan data but removed the line mentioning the range in the description. We have included that the scans were taken between 33% and 50% image overlap on line 6.

Line 332 refers to "over two years" of use, whereas your response letter dates the working version to May 2025. The response letter also mentions that a second laboratory has built TIM from your design files and uses it for research. That is a good answer to Reviewer 1's adoption-readiness concern and you could add something on that subject in the suppl mat.(and you could add this in the new suppl. section I requested above, for instance you could give info on any bugs was encountered and how it was solved. And by the way we do not know how many operators are needed to assemble but also run TIM.

We apologies for the inconsistencies in the timeline. The May 2025 date was when TIM received a major scanning and autofocusing update. Different control software procedures were used to capture images starting in the summer of 2023. We have added a clarification about this on line 329. We have added a note about TIM being recreated by another lab in the supplemental section describing the electronic diagram and assembly instructions.

I could find several typos still in the paper, e.g.;

We thank the editor for pointing out the typos and apologize for them.

Figure 4 is never cited in the text.

We have added a mention of Figure 4 on line 127.

L 271 uses Callitropsis nootkatensis and Table S1 Chamaecyparis nootkatensis; use one name.

Apologies, these are the synonyms for the same species. We have updated to *Callitropsis nootkatensis* throughout as that is the current name.
L 187: "gynosperms".

Done. Corrected spelling to be gymnosperms.

Fig. 5 "due significant scratches"; please also name the species.

Done. Changed to 'due to significant scratches' and we added the name of the species (*Callitropsis nootkatensis*).

L 195- 200 describe the skip rule as "at least 95% of" the previous sharpness in one place and "lower than" the previous neighbour in another.

We removed the sentence mentioning 'lower than' and added more clarity that the system autofocuses when the skip rule does not apply on line 198.

Some references are not completed (e.g. Mansur (2022)).

Apologies. We have now changed the bibliography file in our Latex project to force the writing of the URL to the stitching Python package.

The sentences on LLM need to be in its own section.

We moved the sentences on LLM into its own section and moved it to be below the conclusion on line 376.